

Plasma-Based Sensing Approaches for Aero Environment Characterization

Dr. George Papadopoulos

Site Leader · GE Aerospace - Innoveering



DATE

November 12, 2026

TIME

2:00pm – 3:00pm

LOCATION

ME Conference Rm – ST254

ABSTRACT

The presentation will discuss the application of electrical plasma across the voltage-current spectrum as a foundational approach towards developing novel sensing and processing methods for dynamic flow measurement and characterization. In addition, the use of high current electrical breakdown to generate shock waves traveling in a tube provides a flow simulator for the dynamic calibration of sensors, primarily pressure. It differs from the traditional approach as it provides a closer representation of impulsive disturbances that are dynamically closer to real flow perturbations, while the inherent high repetition rate of shock waver generation uniquely addresses statistical uncertainty requirements.

SPEAKER BIO

Dr. George Papadopoulos has over 35 years of experience in thermal-fluids engineering, measurement science, test engineering, and advanced system development. For the past 25 years, he has worked in the aerospace and defense sector, leading projects in sensing for harsh environments, autonomous embedded systems, and hypersonic test and evaluation. He previously served as Vice President for Advanced Systems and Controls and as Director of Test Services and Advanced Systems Engineering at Innoveering LLC, prior to its acquisition by GE Aerospace in 2022. He is currently Site Leader for Innoveering, overseeing operations and program and engineering execution across multiple locations in New York and Florida.

Dr. Papadopoulos is a Fellow of ASME and a recipient of the ASME Dedicated Service Award. He has held several leadership roles within ASME and is also active in AIAA. He has authored or co-authored more than 100 publications and several patents. He received his Ph.D. in Aeronautics and Astronautics from Polytechnic University in Brooklyn, New York.

Department of Mechanical Engineering

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All welcome · Light refreshments served



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